

MATH 102 Calculus II
Midterm exam (mock)

11 problems, 10 points each

Total: 110 points

15:30 – 18:30, October 17, 2025

1. (10 points) Let $v = \langle a_1, a_2, a_3 \rangle$, $w = \langle b_1, b_2, b_3 \rangle$ be two vectors in $\mathbb{R}^3$. Show that

$$v \cdot w = \|v\| \|w\| \cos \theta$$

where θ is the angle between v and w .

2. (10 points) Let $D, E \subseteq \mathbb{R}^n$. Prove the following:
- (a) (5 points) If D and E are open, then $D \cap E$ is open.
 - (b) (5 points) If $D \subseteq E$, then $\text{Int}D \subseteq \text{Int}E$. ($\text{Int}D$ denotes the interior of D .)
3. (10 points) Consider the function $f(X) = 1/\|X\|$ in the set D in $\mathbb{R}^n$ where $\|X\| \geq 2$. Is the function f uniformly continuous on D ?
4. (10 points) Let f be given as follows:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Determine whether f is differentiable at $(0, 0)$.

5. (10 points) Suppose $u(x, y)$ is a C^2 function from $\mathbb{R}^2$ to $\mathbb{R}$. After a change to polar coordinates $u(x, y) = u(r \cos \theta, r \sin \theta)$, use the Chain Rule to show that

$$u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}$$

6. (10 points) Suppose a differentiable vector function $V(X, t)$, that is, three components depending on the four variables of space and time, gives the velocity of fluid particles moving in $\mathbb{R}^3$, so that

$$\frac{dX}{dt} = V(X(t), t)$$

for one of the particles whose position is given by $X(t)$ at time t .

- (a) (3 points) Derive the acceleration of the particle when $X(t) = C_1 + C_2/t$, $V(X, t) = (C_1 - X)/t$ where C_1 and C_2 are constant vectors.
- (b) (7 points) Prove that in the general cases, the acceleration is given by:

$$A(t) = \frac{d}{dt}V(X(t), t) = \frac{\partial V}{\partial t} + (V \cdot \nabla)V$$

7. (10 points) Let the function f be defined as follows:

$$f(x, y) = e^{x^2+y} \sin(xy)$$

- (a) (6 points) Find the second-order Taylor polynomial $T_2(x, y)$ of $f(x, y)$ centered at $(0, 0)$.
- (b) (4 points) Estimate the error term $R_2(x, y)$ near $(0, 0)$.
8. (10 points) Consider the set S of points (x, y, z) in $\mathbb{R}^3$ space that satisfy

$$z^2 = x^2 + 2y^2 + 1$$

- Let $(a, b, 0)$ be a point in the $z = 0$ plane. Find the local extrema of a function of (x, y) to determine the points of S that are closest to $(a, b, 0)$.
9. (10 points) Find the maximum value and the minimum value of xyz with the constraint $x^2 + y^2 + 2z^2 \leq 4$.
10. (10 points) Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by:

$$f(x, y) = (x + y + \sin(x) \sin(y), y + x \sin(y))$$

- (a) (3 points) Show that f is locally invertible around the point $(x, y) = (0, 0)$.
- (b) (3 points) Find the linear approximation, $L(u, v)$, for the inverse function $f^{-1}(u, v)$ near the point $f(0, 0)$.
- (c) (4 points) Check whether the function is globally invertible or not.
11. (10 points) Consider two functions

$$\begin{aligned} f_1(x, y, z) &= x^2 + y^2 + z^2 - 1 \\ f_2(x, y, z) &= x + y + z - 1 \end{aligned}$$

and the curve

$$C = \{(x, y, z) \in \mathbb{R}^3 : f_1(x, y, z) = 0, f_2(x, y, z) = 0\}$$

- (a) (4 points) For $F = (f_1, f_2)$, prove that C can be parametrized in some small neighborhood of $P = (0, 1, 0)$.
- (b) (6 points) Find the tangent vector of $P = (0, 1, 0)$ on C .

This is the end of the test.

Even if you get a low score on this test, you don't need to feel depressed. As you know, this exam is just preparation for the real one. Identify your weaknesses, and good luck on your midterm exams!